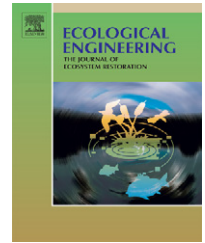


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Agroforestry systems for the production of woody biomass for energy transformation purposes

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ABSTRACT

In the temperate zone, agroforestry systems come increasingly into focus as they offer an approach for the production of fuelwood, thus matching the increasing demand for a self-supply with bioenergy in rural decentralized areas. Because of the large area potential of marginal land, research activities aimed at a reliable estimation of the minimum productivity of fast-growing tree species under most unfavourable site conditions.

Two agroforestry systems were established on reclaimed mine sites in NE-Germany (Lusatia) and Central Germany (Helmstedt). The yield potential and the sustainability of yields were studied for different clones of poplar (*Populus* spp.), willow (*Salix viminalis* L.), and black locust (*Robinia pseudoacacia* L.), considering different rotation periods (3-, 6-, and 9-year-rotation) and approaches of soil amelioration (mineral fertiliser, compost).

On both sites the highest yields of woody biomass were found for *R. pseudoacacia* L. regardless of rotation period and amelioration measures. On loamy substrates in the Helmstedt mining district, all tree species and clones responded positively to soil amelioration measures.

In the agroforestry system in Lusatia, special emphasis was given to the interaction between trees (*R. pseudoacacia*) and crops (*Medicago sativa* L.). Considering the land equivalent ratio (LER), *R. pseudoacacia* hedgerows have practically no negative influence on yields of *M. sativa*. Hence, with regard to an increasing demand for woody biomass, alley cropping with *R. pseudoacacia* and crops such as *M. sativa* may provide a promising alternative for future land use in the temperate zone.

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1. Introduction

Besides minimising or avoiding the utilisation of fossil energy carriers, the enhanced use of renewable energy sources is considered to contribute highly to the reduction of the worldwide CO₂ emissions into the atmosphere and to the protection of

fossil resources (Wisniewski et al., 1993; Hall, 1997). Comparing various options of bioenergy production, it is concluded that the production and use of woody biomass for energy transformation purposes entails numerous beneficial impacts, such as job creation and further positive implications for the added value at regional scales (Hall, 1997; Muschler and Bonnemann,

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1997; Huettl et al., 2000; Volk et al., 2004). In this context, the alternative use of marginal land for the production of woody biomass has been widely discussed (Jug, 1997; Bungart and Huettl, 2002; Volk et al., 2004). Recently agroforestry systems such as alley cropping have come into focus as they integrate the production of lignocelluloses and crops (Gruenewald, 2005; Brandt et al., 2005), thus offering an opportunity for an energetic self-supply in rural areas and also for a diversification of the agricultural production focussing rather on the provision of biomass for energy and industry than on food.

In industrialized countries a high share of marginal agricultural land including post-mining areas are currently not being managed as to the inefficient production and compensatory subsidies still given by the European Union (EU) for leaving these areas untouched. However, the upcoming significant reduction of EU subsidies in the agricultural sector until 2013 calls for innovative, i.e. economically feasible, socially accepted, and ecologically sound strategies for the revalidation of such marginal lands. Under tropical and sub-tropical conditions agroforestry systems have proven to cope with these aims (Kang, 1997; Muschler and Bonnemann, 1997). Thus, also for temperate regions the integration of trees in the agricultural landscape, particularly on marginal arable land, is supposed to be a promising option for future land use. However, with regard to the selection of trees for such land use systems there is still a lack of knowledge in terms of the yield potentials and the yield sustainability of different fast growing tree species under unfavourable conditions.

In order to contribute to the discussion about a sustainable use of marginal land, two field experiments were established in former lignite mining areas. One field experiment refers to the establishment of an alley cropping system in NE-Germany (Lusatian lignite mining district) in 1996. At this site overburden sediments consist of quaternary substrates. In 2002, a second field trial was established in central Germany (Helmstedt lignite mining district) representing dump substrates derived from both quaternary and tertiary overburden sediments. Tertiary substrates are mixed up with lignite and contain pyrite (FeS_2) in various amounts. Both field experiments cover the main types of substrates typical for post-mining landscapes of lignite mining, and, therefore, are supposed to provide representative data on yield potentials and the yield sustainability of different fast growing tree species under most unfavourable edaphic and climatic conditions.

The present study will inform about: (1) the suitability of different tree species for marginal sites in terms of their yield

potential and the sustainability of biomass supply for energy transformation purposes, (2) the influence of different management options (rotation period, soil amelioration) on the productivity of fast growing tree species, (3) the influence of trees on associated crops in agroforestry systems, and (4) an assessment of alley cropping systems as an option for the future use of marginal sites.

2. Materials and methods

2.1. Experimental sites

The main characteristics of the experimental sites under study are given in Table 1. At both sites substrates derived from overburden sediments dumped in the course of opencast lignite mining. In the Lusatian lignite-mining district a total of about 77,000 ha are currently affected by ongoing mining activities (Drebenstedt, 1998). In the Helmstedt lignite mining district the reclamation area comprises 2800 ha of former agricultural land (Pflug, 1998).

The average temperature is between 8 °C (Helmstedt) and 9 °C (Lusatia). The mean annual precipitation of the two sites varies between 569 mm in Lusatia and 623 mm in Helmstedt (Table 1).

The initial site mapping at Lusatia resulted in seven substrate types dominated by loamy sand and sandy loam. The main substrate type in Helmstedt is considered as loam (classification according to USDA). N contents in Helmstedt were higher as compared to Lusatia. Further details of the experimental sites are given by Brandt et al. (2005) and Gruenewald (2005).

2.2. Tree species/clones and crops

In order to cover the range of potentially suitable fast growing tree species different clones of poplar (*Populus* spp.), willow (*Salix viminalis* L.) and black locust (*Robinia pseudoacacia* L.) were selected for the establishment of the agroforestry systems at both experimental sites. Although at that time black locust had not been tested before in such an experimental context its adaptation to marginal sites conditions was common knowledge. Poplar clones and willow were included due to positive experiences made on former arable land (Hofmann, 1995). At both sites two balsam poplar clones (*Populus maximowiczii* Henry $\times$ *Populus trichocarpa* Torr. et Gray: Androscoggin and Hybride 275) and black locust (*R. pseudoacacia* L.) were

Table 1 – Description of the experimental sites

Feature	Experimental agroforestry system	
	Lusatia	Helmstedt
Region/state	NE-Germany/Brandenburg	Central Germany/Lower Saxony
Elevation (m asl)	65	108
Aspect	N, S slopes and valley bottom	Plain
Temperature (°C)	9.4	8.3
Precipitation (mm)	569	623
Dominant substrate	Loamy sand/sandy loam	Loam
N _t , 0–20 cm (%)	<0.01	0.14

Table 2 – Description of the experiments

Feature	Experimental agroforestry system	
	Lusatia	Helmstedt
Establishment	1996	2002
Area (ha)	4	2
No. of test species	4 tree species/clones (A ^a , H ^b , R ^c , S ^d), 1 crop ^e	4 tree species/clones (A ^a , B ^f , H ^b , R ^c), 1 crop ^e
Size of tree plots (m)	6.1 × 10.2	7.25 × 60 (<i>R. pseudoacacia</i>) 2.75 × 60 (<i>Populus</i> spp.)
Spacing of trees (m)	Inter-row: 0.75 × 0.6 Between double rows: 1.6 Between hedgerows: 21	Inter-row: 0.75 × 0.6 Between double rows: 2.0 Between hedgerows: 51
Width of alley (m)	18	51
Rotation periods (a)	3	–
	6	
	9	
Fertiliser treatment	–	Compost Mineral fertiliser Compost + mineral fertiliser
No. of replicates per treatment	12	2
Years of fertilisation	Crops: 1996–2002 Trees: 1998–2002	2002

^a *P. maximowiczii* Henry × *P. trichocarpa* Torr. et Gray (Androscoggin).

^b *P. maximowiczii* Henry × *P. trichocarpa* Torr. et Gray (Hybride 275).

^c *R. pseudoacacia* L.

^d *S. viminalis* L. (Carmen).

^e *M. sativa* L.

^f *P. trichocarpa* Torr. et Gray × *P. deltoides* Bartr. (Beaupré).

tested (Table 2). First studies showed that willow (*S. viminalis* L. cv. Carmen) is not adapted to the prevailing climatic and edaphic conditions (Gruenewald, 2005; Bungart and Huettl, 2004). Therefore, the poplar clone Beaupré (*P. trichocarpa* Torr. et Gray × *Populus deltoides* Bartr.) was chosen instead for the field trial at Helmstedt. At this site black locust was planted 1 year after the poplar as seedlings with local proveniences were missing. The poplar and willow clones were planted as cuttings, the black locust as 1-year-old rooted transplants.

Trees at Lusatia were planted in double rows at spacing of 1.6 m between the double rows, 0.75 m in between and inter-row distances of 0.6 m. Buffer strips (width 1.5 m) on each side of the hedgerows were kept unmanaged in order to minimize root competition between trees and crops (Fig. 1b). Tree plots were 6.1 m × 10.2 m in size.

At Helmstedt trees were planted in double rows at a 2-m spacing between the double rows, 0.75 m in between rows and inter-row distances of 0.6 m. Tree plots were 7.25 m × 60 m in size corresponding to three double rows of trees. On a plot level each double row was representing a different poplar clone. So sub-plots of one poplar clone were 2.75 m × 60 m in size (Fig. 2).

For the field trial in the Lusatian mining district crops were grown in 18 m wide alleys between tree hedgerows (Fig. 1a). A 1.5 m buffer strip between hedgerows and crops was left unmanaged on both sides of the alley. The crop rotation was as follows: summer rye (1996), winter rye (1996/1997), lupine (1997), winter rye (1997/1998), fallow (1998/1999), alfalfa

(1999–2002). However, data presented in this paper only refer to the period from spring 1999 to summer 2002 when only alfalfa (*Medicago sativa* L.) was cultivated. At Helmstedt alfalfa was cultivated between tree hedges on a field 200 m × 51 m in size (Fig. 2).

2.3. Management practices

The experimental site in Lusatia was established in April 1996 after an initial application of 15 t ha⁻¹ lime (41% CaO) and N/P/K fertilizer (100/100/100 kg ha⁻¹) in autumn 1995. Between 1996 and 1998 N/P/K fertilizer was applied in alleys applied at various amounts cumulating to about 220/190/240 kg ha⁻¹. In time period referred to in the present study alfalfa was fertilized annually, but only initially nitrogen was included (1999: 30/20/85 kg ha⁻¹, 2000: 0/25/100 kg ha⁻¹, 2001: 0/25/95 kg ha⁻¹, 2002: 0/30/175 kg ha⁻¹).

N/P/K-fertiliser was applied to hedgerows in 1998 (30/100/60 kg ha⁻¹), 1999 (0/10/60 kg ha⁻¹), 2000 (20/100/0 kg ha⁻¹), 2001 (20/0/0 kg ha⁻¹) and 2002 (30/100/60 kg ha⁻¹) in order to compensate for nutritional imbalances as detected by leave analysis.

Treatments at Lusatia included rotation periods of 3, 6 and 9 years with 12 replications each in a randomized design (Table 2).

The experimental site at Helmstedt was ameliorated with 116 t CaO ha⁻¹ before planting in 2002 to compensate for the

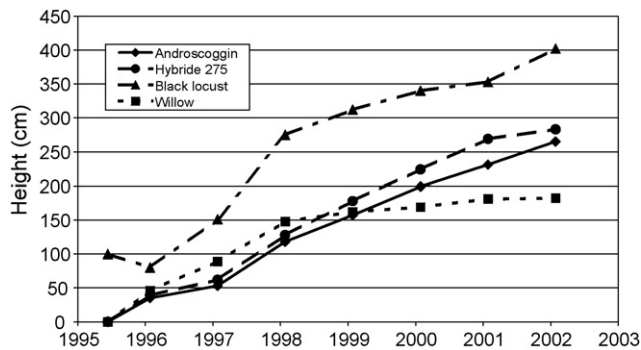


Fig. 3 – Height increments of tree species and clones, respectively, at the Lusatian study site; height measurements were carried out in December each year.

constant increments. Therefore, height growth data already reflect a predominant growth performance of *R. pseudoacacia* under the given climatic and edaphic regime. Hence, *R. pseudoacacia* appeared to be much more adapted to those specific conditions.

At the Lusatian study site the above-ground biomass accumulation after 6 years ranged from 4.5 t ha^{-1} dry matter for *S. viminalis* to 29.8 t ha^{-1} dry matter for *R. pseudoacacia* (Fig. 4a). This enormous variation in productivity reflects differences of the adaptability of *Populus* spp., *S. viminalis* and *R. pseudoacacia* to specific site conditions even more clearly than height

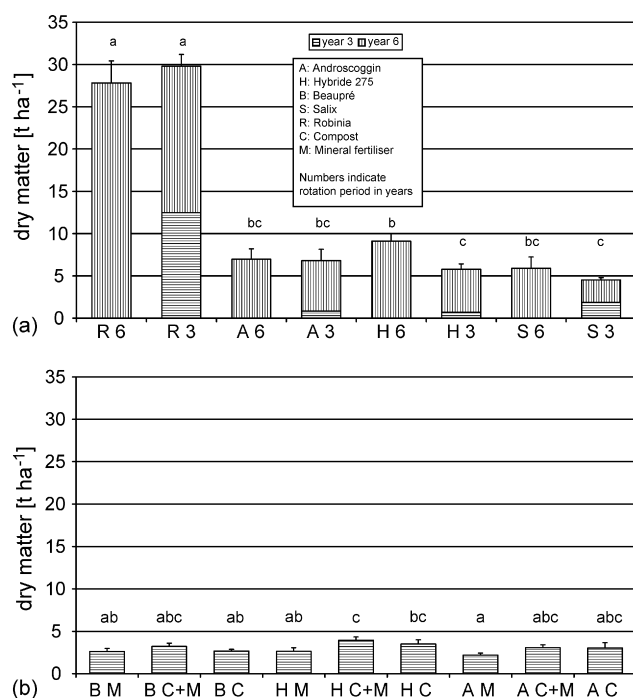


Fig. 4 – (a) Average above-ground biomass accumulation for different tree species/clones and rotation periods of 3 and 6 years at Lusatia; bars indicate standard errors. (b) Average above-ground biomass accumulation for different soil amendments after a 3-year-rotation at Helmstedt; bars indicate standard errors.

does. The rotation period had no significant effect on tree biomass accumulation except for poplar clone Hybride 275. With regard to the annual biomass productivity, highest values of *S. viminalis* were achieved after a 6-year-rotation with $1.0 \text{ t ha}^{-1} \text{ a}^{-1}$. For *Populus* spp. and *R. pseudoacacia*, biomass productivity peaked in the second rotation period of the 3-year-rotation to 2.0 and $5.8 \text{ t ha}^{-1} \text{ a}^{-1}$, respectively.

After the first 3-year-rotation the average biomass accumulation at the Helmstedt study site varied between 2.2 t ha^{-1} dry matter for clone Androscoggin (mineral fertiliser treatment) and 3.9 t ha^{-1} dry matter for Hybride 275 (mixed application of compost and mineral fertiliser). Data for *R. pseudoacacia* were only available for the first 2 years and are, therefore, not shown in Fig. 4b (see Section 2.2). Yields ranged between 3.2 t ha^{-1} (mineral fertiliser) and 4.6 t ha^{-1} (compost) confirming the outstanding growth performance of that species under marginal conditions.

In general, for *Populus* spp. maximum yields resulted from a mixed application of mineral fertiliser and compost: an average of only $2\text{--}2.6 \text{ t ha}^{-1}$ dry biomass was calculated for poplars after application of mineral fertiliser without compost, whereas, a maximum biomass accumulation of $3\text{--}3.9 \text{ t ha}^{-1}$ was reached when applying mineral fertiliser and compost. The annual productivity of *Populus* spp. at Helmstedt was approximately two times higher as compared to those at the study site Lusatia when only mineral fertiliser was applied. The biomass accumulation was even three to four times higher at Helmstedt for subplots of poplars treated with mineral fertiliser and compost. This points out on the importance of the amendment of organic matter for the productivity on post-mining sites.

Similar studies on former arable land (Hofmann-Schielle et al., 1999) reported about yields of about $5 \text{ t ha}^{-1} \text{ a}^{-1}$ after a 4-year-rotation for *P. trichocarpa* Torr. et Gray. In the same experiment productivity of *Salix* spp. varied between 2 and $4 \text{ t ha}^{-1} \text{ a}^{-1}$. However, yields increased significantly in the second rotation period for both *Populus* spp. and *Salix* spp. peaking to $13 \text{ t ha}^{-1} \text{ a}^{-1}$. A minimum productivity of approximately $5\text{--}6.5 \text{ t ha}^{-1} \text{ a}^{-1}$ was measured for *Populus* spp. on marginal sites in Germany (Weisgerber, 1983; Busch and Kreysa, 1985; Dimitri, 1988). Various studies in the German and Swedish context (Christersson, 1986; Hofmann, 1995; Jug, 1997) showed that the productivity of *Salix* spp. is generally lower than for *Populus* spp. under same site conditions, however, in some cases yields of 10 and $18 \text{ t ha}^{-1} \text{ a}^{-1}$ could be achieved.

Results for the productivity of *R. pseudoacacia* in Lusatia resemble those reported in the literature, ranging from 3.3 to $8 \text{ t ha}^{-1} \text{ a}^{-1}$ (Dickmann et al., 1985; Geyer, 1989; Grassi and Bridgwater, 1991; Bongarten et al., 1992). Despite the low soil fertility and unfavourable climatic conditions prevailing in the Lusatian agroforestry system yields for black locust appeared to be relatively high compared to those found in literature.

According to a definition of Roehrig (1979) fast growing tree species are supposed to exceed 5 t biomass production per hectare and year. *Populus* spp. and *Salix* spp. were found to fall significantly below that threshold value. However, since this threshold value had been defined for high productivity sites, it hardly applies to marginal or low productivity sites. For such sites the maximum productivity of *Populus* spp. and *S. viminalis* ranged between 3.2 t ha^{-1} (poplar clone Androscoggin)

and 6.4 t ha^{-1} (poplar clone Beaupré) after a 6-year rotation (Bungart and Huettl, 2002) under the given continental climate of Lusatia for sandy loams.

Our results indicate that *Populus* spp. is not sufficiently adapted to the nutrient-poor sandy substrates and the low precipitation regime at the Lusatian study site. At Helmstedt conditions seemed to be more favourable for *Populus* spp. because after 3 years biomass increments were much higher than at the Lusatian study site after the first 3 years. These findings emphasise that high biomass increments for poplar can only be achieved on loamy sites or through soil amelioration. However, yields fall still below the threshold of $5 \text{ t ha}^{-1} \text{ a}^{-1}$ (shown above). The positive influence of fertiliser and compost amendments on biomass production at the Helmstedt study site, therefore, reflects the limitations of nutrient and water supply possibly existing at the reference site in Lusatia.

Based on our results and studies of Bungart and Huettl (2002) *S. viminalis* is not adapted to such site conditions at all, and, therefore, should not be considered furthermore.

Annual yields of *R. pseudoacacia* matched or even exceeded the yields reported in the literature. Because of differences in site conditions and planting densities results of different sites cannot simply be compared. However, present results clearly show that *R. pseudoacacia* is well adapted to nutrient-poor sandy substrates and the low precipitation regime.

3.2. Ash content, ash softening temperature, and calorific value of wood

Table 3 shows the ash content, ash softening temperature, and calorific value of the wood for the tree species and clones under study. The calorific value in terms of energy content per mass unit was not significantly different between tree species. However, due to the high wood density of *R. pseudoacacia*, the calorific value expressed in terms of MJ per cubic meter of wood chips (Table 3) is significantly higher for *R. pseudoacacia* than for poplars and willow. This will be economically relevant if storage or transport capacity is a limiting factor. Based on the annual productivity of the second 3-year-rotation period and, taking into consideration the survival rate the annual energy productivity of *R. pseudoacacia* is at least more than three times higher than any other tree species under study. A further advantage refers to the very low ash content of *R. pseudoacacia* of about 1.5%, which makes the wood highly suitable for combustion processes. However, the melting behaviour reflected by the ash soft-

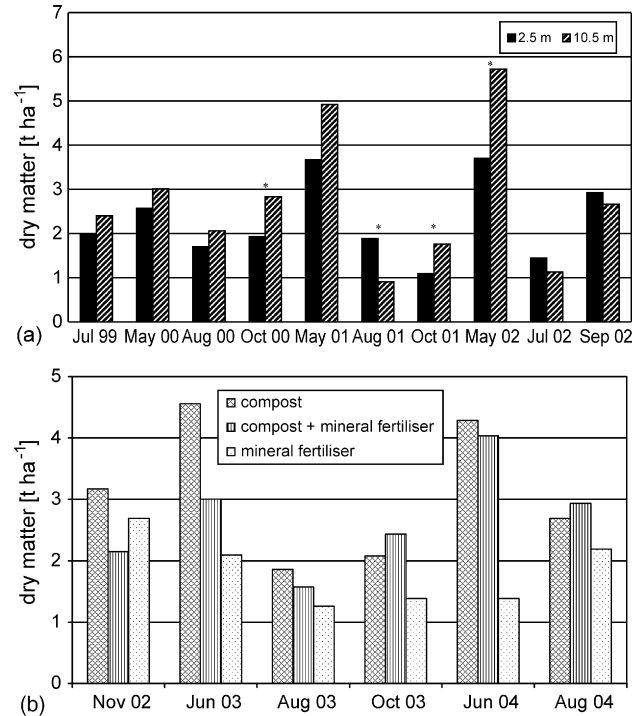


Fig. 5 – (a) Yield of *M. sativa* on the Lusatian study site as a function of the distance to the *R. pseudoacacia* hedgerow. Significant differences between yields from different distances to the tree hedgerow are marked by an asterisk. (b) Yield of *M. sativa* on the Helmstedt study site as a function of soil amendments.

ening temperature does not differ significantly between tree species.

3.3. Yield performance of *M. sativa* L.

The annual yield of *M. sativa* at the Lusatian study site ranged between 7.6 and $9.5 \text{ t ha}^{-1} \text{ a}^{-1}$ dry matter (Fig. 5a) which falls below the mean annual productivity on naturally grown arable land in Germany ranging between 9 and $14 \text{ t ha}^{-1} \text{ a}^{-1}$ dry matter (Heyland, 1996). However, according to experiences of local farmers *M. sativa* yields achieved on these reclaimed overburden sediments nearly equalled that of adjacent agricultural sites of the study area. This clearly reflects the generally low nutrient supply and water availability on farm lands in this region. Therefore, it was even more surprising that crop yields developed in such a positive way 7 years after the reclamation of the overburden sediments.

Table 3 – Energy related parameter of wood from Lusatian site

	Ash content (mass%)	Ash softening temperature ($^{\circ}\text{C}$)	Calorific value (kJ kg^{-1})	Calorific value (MJ m^{-3} wood chips)	Calorific value ($\text{MJ ha}^{-1} \text{ a}^{-1}$) ^a
Androskoggin	1.6	1552	17,550	2886	32,687
Hybride 275	2.1	1566	17,430	2854	25,492
Black locust	1.5	1520	16,540	4818	113,216
Willow	2.2	1544	17,410	3619	14,447

^a Based on results of the second rotation period of the 3-year-rotation treatment.

At Helmstedt the accumulated biomass of alfalfa varied between 4.7 t ha^{-1} dry matter in the mineral fertiliser treatment and a maximum of 8.5 t ha^{-1} dry matter, for compost treatments. Yields of 7.0 t ha^{-1} dry matter were attained for mixtures of mineral fertiliser and compost (Fig. 5b). These results correspond with the positive growth response of most tree species for same treatments and underline the significance of organic matter amendments for the productivity of sites.

In Fig. 5a yields of *M. sativa* are shown as a function of distance to the hedgerows of *R. pseudoacacia*. Significant differences between the yield of *M. sativa* growing in 2.5 m distance to *R. pseudoacacia* hedgerows and yields calculated from samplings in the middle of the alley (10.5 m distance) without any above- or below-ground interference of *R. pseudoacacia* were only observed at four out of 10 harvests. In one case the yield of *M. sativa* growing next to trees was even higher than in the centre of the alleys. But in all other cases the yield of *M. sativa* growing next to the hedgerows was decreased as compared to the centre of the alleys. To compare the productivity of *M. sativa* with and without consideration of such edge effects weighted averages of the yield of alfalfa were calculated with the assumption that 11% of the plot was influenced by above- and below-ground interactions of trees and crops. Cumulated yield of all 10 harvests based on these weighted averages amounts to 26.7 t ha^{-1} . Based on yields from the middle of the alley the cumulated yield of 27.4 t ha^{-1} clearly indicated that potential edge effects will not negatively influence the overall growth performance of crops.

3.4. Land equivalent ratio of the alley cropping system in Lusatia

For comprehensive validation of the productivity of the alley cropping system at the Lusatian study site the land equivalent ratio (LER) was calculated. LER is defined as the land amount equivalent needed by one type of crop to break even in physical terms (i.e. t ha^{-1}) with what is yielded of 1 ha by another (Mead and Willey, 1980). Thus, the LER allows to estimate the surface area needed by monocropping to obtain the same production than the experimental alley cropping system with its tree and crop components. Hence, a minimum LER for the alley cropping system should not be below 1, which would reflect a positive synergistic or at least indifferent mutual relationship between trees and crops (Lal, 1991). For the calculation of LER, a time of 3 years was considered including nine harvests of *M. sativa* and one harvest of *R. pseudoacacia* after a 3-year-rotation.

The slight decrease of the yield of alfalfa in the alley cropping system as compared to monocropping (see Section 3.3) leads to a LER of 0.98 for the alley cropping system. Hence, under the given climatic and edaphic conditions of the Lusatian post-mining landscapes the productivity of alfalfa for an agroforestry system and for monocropping do not differ significantly which means that an area of approximately 1 ha is needed to obtain the same yield irrespective of the type of land use applied. Thus, it is confirmed that the mutual relationship between *R. pseudoacacia* and *M. sativa* was almost indifferent.

The LER of agroforestry systems must not always match the level of traditional land use systems. However, social, ecologic, and other non-monetary benefits resulting from such sys-

tems need to be considered, as well. Important topics in this context are the quality of seepage water, the increase of biodiversity, carbon sequestration or groundwater production and aesthetic values. Furthermore, it is strongly recommended to base calculations on a longer period including a rotation of different crops and several harvests of trees.

3.5. Root distribution at the experimental site Lusatia

Since no significant effects of *R. pseudoacacia* on yields of *M. sativa* could be found, it was hypothesized that in that part of the alley were below-ground interactions between both species occur trees and crops rather use the nutrient and water resources in a complementary way by extending root growth in different soil layers.

A soil profile, which was applied in a right angle to hedgerows of *R. pseudoacacia* to a depth of 1 m, provided a visual assessment of the horizontal growth of roots from *R. pseudoacacia*. Roots of *R. pseudoacacia* were still present in the alleys within a distance of 2.5 m from the hedgerow but did not occur beyond a distance of 4 m. These findings supported the findings made above (see Section 3.3) with regard to the influence of hedgerows on yields of *M. sativa* within a distance of 2.5 m. Based on this information sampling points for the estimation of the root distribution were defined as follows: “Black locust” samples were taken in the middle of *R. pseudoacacia* (black locust) plots. Only roots of black locust occur. “Alfalfa (2.5 m)” samples were taken from the alleys 2.5 m far from the hedgerow. Below-ground both roots of alfalfa and black locust occur. “Alfalfa (10.5 m)” samples were taken from the middle of the alley. Only roots of *M. sativa* (alfalfa) occur.

With regard to all sampling dates the root length density of *M. sativa* is evenly distributed throughout the sampled profile from 0 to 45 cm depth (Fig. 6). On the other hand root length distribution under *R. pseudoacacia* shows a clear vertical gradient with values decreasing from 5.9 cm cm^{-3} at the surface to values below 1 cm cm^{-3} in 30–45 cm depth.

M. sativa is a perennial crop and is well known for extending its roots growth even below the ploughing horizon. Benzarti (1999) detected roots of *M. sativa* still in 0.8 m depth with most roots growing in 0–30 cm depth. This is confirmed by our own results where roots of alfalfa were found down to 0.9 m depth. For nutrient poor sites Fox and Lipps (1955) reported *M. sativa* roots extending to 2 m depth for accessing subsoil nutrient and water pools. The observed rooting patterns of *M. sativa* reflect the high ameliorative benefit of this species with regard to soil structure formation and the adaptive potential with regard to drought.

The shallow root proliferation of *R. pseudoacacia* does not appear to provide an ameliorative effect with regard to the exploration of nutrient and water pools in deeper soil layers as being expected for trees in agroforestry systems. Surprisingly this did not affect the growth performance of *R. pseudoacacia*. Therefore, apart from root distribution other adaptive mechanisms appear to exist that allow trees to quickly and effectively access even low amounts of precipitated water and fertiliser. Recent findings emphasize the role of the nutrient and water use efficiency, respectively, for the growth performance of fast growing tree species on clayey-sandy mine sites (Bungart et al., 2001; Bungart and Huettl, 2004).

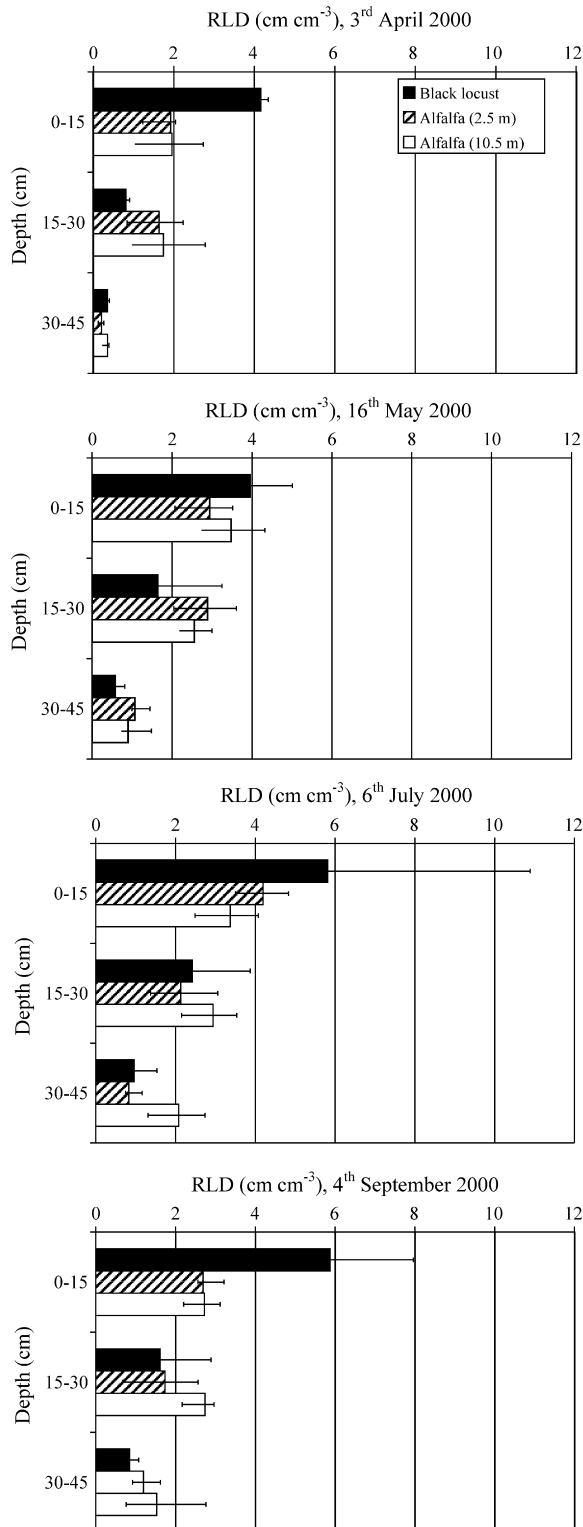


Fig. 6 – Root length density (RLD) under *R. pseudoacacia* and under alfalfa with a distance of 2.5 and 10.5 m to the hedgerow at four different dates during the vegetation period of the year; bars represent median values, error bars show interquartile range.

Considering the desirable complementary use of water and nutrient resources in agroforestry systems by trees and crops van Noordwijk et al. (1996) stated that a nutrient uptake from deeper horizons only occurs if at least a limited resource is available in the subsoil. On the experimental site the prevailing precipitation regime does not allow an extended infiltration, retention and storage of water in the subsoil. The availability of nutrients is limited to the application of fertiliser generally stimulating root growth in the topsoil. Hence, low water and nutrient resources do not favour root proliferation in deeper soil layers. Thus, the coinciding root growth and the periodical occurrence of competition between *R. pseudoacacia* and *M. sativa* at the Lusatian study site may be well explained.

With regard to the combination of an N-fixing species with a non-N-fixing species, the observed intermingling root system of trees and crops can be advantageous. Such a rooting pattern favours that N provided by the N-fixing species can be used by the non-N-fixing species (Schroth, 1999).

4. Conclusions

In this investigation special emphasis was placed on the assessment of the sustainability of biomass production of fast growing tree species on marginal land.

The results show that a sustainable supply of fuelwood is possible even under marginal conditions if tree species well adapted to specific climatic and edaphic conditions are selected. Highest productivity was achieved by *R. pseudoacacia* regardless of site conditions and amelioration measures. The maximum annual productivity for this species was 5.8 t ha⁻¹. Growth performance of *Populus* spp. may be same under improved site conditions. This was indicated by increased yields after amelioration of loamy coal containing substrates with compost. However, with regard to the lifespan of agroforestry systems more long-term investigations are needed to verify the sustainability of the biomass production.

Results of this study show the significance and potentials of soil amendments for the site productivity and tree species selection. Productivity of poplar clones could be increased by approximately 30% through the application of compost and mineral fertiliser. With regard to rotation period no clear influence on the yield of fast growing tree species could be found except for poplar clone Hybride 275.

Only a minor influence of *R. pseudoacacia* hedgerows on yield of *M. sativa* was observed although root systems of both *M. sativa* and *R. pseudoacacia* competed for same resources in the topsoil layer. It is concluded that the combination of well adapted tree species and crops in an agroforestry systems offers a promising alternative to farmers for integrating fuelwood and crop production within one management system and, parallelly, enhancing site fertility through positive ameliorative effects from nitrogen fixation and soil structure formation by root proliferation.

The land equivalent ratio (LER) for an alley cropping system with *R. pseudoacacia* and *M. sativa* at the Lusatian study site proved potential above- and below-ground interactions between trees and crops to remain indifferent. The low LER also indicates that more research efforts are needed to increase the economic competitiveness of such systems under

the given conditions. At the same time future research activities should place more emphasis on ecological, social and other non-monetary benefits and their economic relevance.

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